REACTION MECHANISM DIGITAL TWIN

A vendor-neutral R&D baseline for pathway evidence, transition states and kinetics, built with Vicena

OPEN SOURCE

MIT • Commercial use allowed

Turn structures and saved quantum outputs into an inspectable mechanism workflow.

Validate metadata, prepare endpoints, interrogate transition states, preserve failures, and compare calculated evidence with experiment.

INPUT READY

Mapped structures • charge • spin
Units • conditions • provenance

ENDPOINT MODELING

Local geometries • Rowan conformers
Saved UUIDs • raw outputs

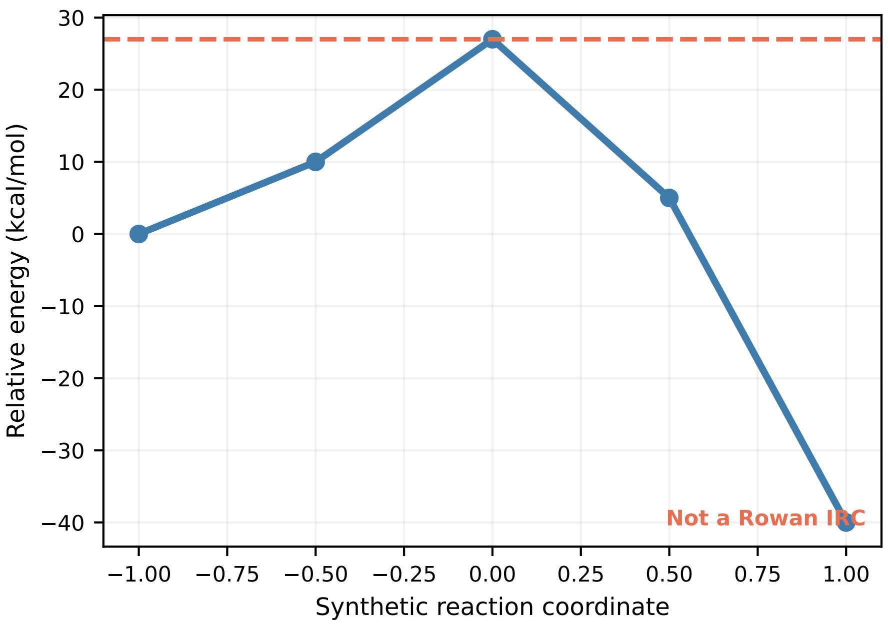
TS ACCEPTANCE

Imaginary mode • geometry • mapping
IRC or equivalent path evidence

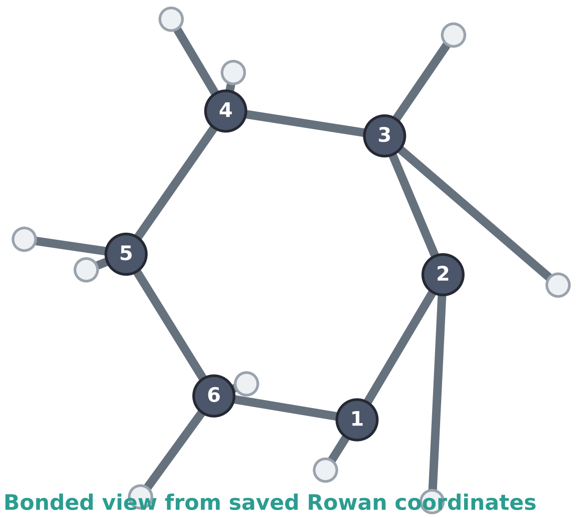
AGENT ADAPTABLE

Schemas • playbooks • tests
New reactions without rewrites

Archived energy logic fixture



Rowan product conformer



1 completed
Rowan conformer workflow

0.37 credits
Provider charge

1 conformer
Saved endpoint geometry

12 tests
Repository baseline checks

ONE PLATFORM, AN AUDITABLE MECHANISM WORKFLOW

DEFINE
structures + metadata

PREPARE
endpoint conformers

SEARCH
transition-state guess

VALIDATE
frequency + path

COMPARE
experiment + uncertainty

github.com/vicena-labs/reaction-mechanism-digital-twin

vicena.ai • Validate the path before trusting the barrier